

Do pre-literate children spontaneously employ spatial coding for serial order in working memory?

Jean-Philippe van Dijck^{1,2}, Elger Abrahamse³ & Wim Fias²

¹Department of Applied Psychology, Thomas More, Antwerp, Belgium

²Department of Experimental Psychology, Ghent University, Ghent, Belgium

³Department of Communication and Cognition, Tilburg University, Tilburg, the Netherlands

Correspondence to:

Jean-Philippe van Dijck

Thomas More University of Applied Sciences

Department of Applied Psychology

Molenstraat 8, 2018 Antwerp, Belgium

e-mail: jean-philippe.vandijck@thomasmore.be

ABSTRACT

The ability to memorize arbitrary sequences contributes to cognitive faculties like language and mathematics. Research suggests that in literate adults, serial order in verbal working memory (WM) is grounded in spatial attention and is mentally organized according to our reading habits, i.e. from left-to-right in Western cultures. Currently, it is unknown whether spatialization is a consequence of literacy, or whether the ability already exist early in life but is shaped by literacy in ‘calibrating’ the initial individual differences in the orientation of spatial coding. Here, we investigated the spatial coding of serial order in WM in 5-year old children who did not yet enter formal literacy education. At the group-level, no systematic spatial coding was observed. To investigate whether this absence was due to subjects with reliable but opposing effects, we determined the prevalence of spatial coding at the individual level. This analysis revealed that 36% of the children systematically associated serial order to space, with approximately half of them coding from left-to-right and the rest from right-to-left. These results indicate that a subgroup of pre-literate children associate serial order with space and suggest that reading and writing experience ‘calibrates’ the orientation of spatial coding with our reading habits.

Keywords: serial order, working memory, space, children, development

INTRODUCTION

The ability to maintain the order of items in working memory (WM) is an important ability for many higher-order cognitive functions such as reading, mathematics, and reasoning. Therefore, much effort has been devoted to understanding the mechanisms underlying this ability. Currently, the dominant model holds that serial order in WM is achieved via position marking. Position marking has its origins in computational and theoretical modelling (for a review see ref. 1) and refers to the notion that conjunctions between memoranda and a pre-existing mental reference frame serve to encode and maintain ordinal information ².

Several candidate types of position markers have been proposed such as time signals ³ and magnitude codes ⁴. Recently, spatial coding was proposed on the basis of systematic interactions between serial order (of non-spatially defined items) in WM and space-related responses. Evidence for this idea came from a study by van Dijck and Fias ⁵. They asked participants to store sequences of fruits and vegetables in verbal WM. During the retention interval, positive and negative probe words were presented. When the probe item belonged to the sequence, participants had to categorize the word as fruit or vegetable by pressing a left or a right button. They found that items from the beginning of the sequence were faster categorized with left-sided responses (compared to right ones), while the reversed was observed for end items. This observation is called the Ordinal Position Effect (OPE) and has been replicated across various experimental settings (e.g., refs. 6–14). Accordingly, the mental whiteboard hypothesis was proposed: When confronted with the challenge of serial order in (non-spatial) WM, our cognitive system spontaneously generates an internal spatial template to which memoranda are bound in a systematic fashion, such that the resulting spatial configuration reflects their serial order ^{15,16}.

To our knowledge, no previous studies have explored at what stage of development the ability to process serial order in WM – and the role of spatially defined position marking therein – emerges. Already from a very young age, children are able to process and learn about serial order ¹⁷. Yet, it remains unknown to what extent spontaneous spatial coding

serves as an underlying mechanism for serial order in WM ^{footnote 1} across the stages of development. A first step is to determine whether spatial coding of serial order exists in children before they enter formal literacy education. Afterall, being a (fluent) reader could be a critical trigger for such a mechanism, as reading and writing provides a natural link between serial order processes and spatial cognition. Guida and colleagues ¹¹ investigated the impact of reading direction in adults and found that literacy is required to show an OPE, and that the orientation of the OPE depends on the reading direction of the tested individuals (i.e., depending on reading direction, the OPE followed a left-to-right vs right-to-left orientation). Yet, the study of Guida and colleagues tested adults rather than children and is therefore not informative to what happens at early ages. So, apart from the possibility of learning to read being the trigger, it is also possible that the tendency to map verbal serial order to (horizontal) space is present before formal literacy education. Here, individual differences in orientation are likely, since cultural conventions about the representation of ordered information is known to be important in determining orientation ¹⁸.

Here, we investigated the OPE in 5-year old children who only received training in pre-literate skills (e.g. letter/symbol detection, story reconstructing using images, flipping books), without dedicated and intensive formal literacy education at school. Besides performing group-level analyses, we also applied a novel statistical technique that can be used to determine the presence of the OPE at the individual level ¹⁹. As such, we can obtain a more detailed view on the prevalence of individual differences in the orientation of the OPE. To our knowledge, this prevalence is not yet investigated in adults. To provide a comparison standard, we reanalysed several of existing datasets on the OPE in adults with different reading habits. These analyses revealed that about 45% of the individuals demonstrate a reliable OPE. The prevalence of left-right and right-left coders is strongly influenced by reading direction, as the majority (i.e. 87.5%) maps serial order according to their reading habits (Appendix 1).

METHODS

¹ Rather than WM, some researchers may argue that we should refer to the precursors of WM since literacy may have a significant role in the development of WM per se (e.g. ref. 42). This issue is orthogonal to the goals of the current paper.

Participants

In total, 84 children from the last year of kindergarten (average 5.07 years; $SD = 0.26$) were tested (50 females and 34 males). 81 children spoke a home language with a left-right reading and writing direction, 2 children spoke a right-left language at home and for 1 child, this information was missing. A test session lasted approximately 25 minutes. Before participation an informed consent was obtained from the parents. This study was in line with the protocols approved by the ethical committee of the Faculty of Psychology and Educational sciences of Ghent University. All children had normal or corrected-to-normal vision. The data of four children were discarded from the analyses because of external distraction during testing, the inability to pay attention to the task or not following the task instructions.

Material and experimental setup

The experiment was conducted on a 15-inch Elo touchscreen which was connected to a laptop which controlled the item presentation using E-Prime. The child was sitting at a viewing distance of approx. 30 cm and had to put their hand at a sticker of a red dot which was stuck 10 cm from the screen. The red dot and the body midline of the child were aligned to the centre of the screen. Responses were given with a hand of choice, and the children were instructed to put this hand on the dot when not responding throughout the experiment. All participants were tested at school during class hours in a quiet room next to the classroom. The experimenter (with laptop) was sitting left or right from the touchscreen.

Memoranda were coloured line drawings of familiar objects. To make sure that the children knew the displayed memoranda (semantic processing of the items is important for the OPE to occur⁹), they were selected on the basis of the age of acquisition of the corresponding word which had to be smaller than or equal to 5 years²⁰. In this way a set of 78 drawings was created. From this set one training sequence and different randomized memory sets (each containing 12 sequences of three items, see below) were generated which were randomly assigned to the participants.

Procedure

The experiment consisted of 12 different experimental blocks and 1 training block. Each block consisted of three subsequent phases: an encoding phase, a probe detection phase and a control phase (see Figure 1). During the encoding phase, 3 drawings (ca. 7.5 by 7.5 cm) were successively presented at the centre of the screen. To ensure successful encoding, the

presentation of the sequence was experimenter-paced, and the children were encouraged to name the object they saw on the image (to stimulate semantic processing) and to repeat it in correct serial order together with the sequence items they already heard. Each sequence was presented twice during the encoding phase and between the two presentations the children were asked to recall the entire sequence in the order of presentation.

In-between the encoding phase and the start of the probe detection phase, a retention interval of 2000ms passed. This interval ended by the presentation of a red dot, which was used as a reminder to put the response hand on the red dot. The experimenter initiated the start of the probe detection phase with a key press. The probe detection phase started with the presentation of a central fixation cross (500ms) after which two items were simultaneously added to the screen (size images 7.5 by 7.5 cm), one presented on the left side, and the other on the right side (the distance from the fixation cross to the midpoint of each image was 11 cm). One of the items was part of the memorized sequence, the other one was a novel item. The children were instructed to touch as fast and accurate as possible the item on the screen which belonged to the memorized sequence. There was no response deadline. After responding, feedback (1500ms) appeared visually (a sad or happy smiley) and auditorily (a buzzer or applause). The intertrial interval was set to 1000ms, and care was taken that the child returned the hand to the red dot after responding. For each sequence, 6 such trials were presented so that each memory item was presented once on the left and once on the right side of the screen. Finally, the control phase started to test whether the child remembered the entire sequence in mind in correct order. For this purpose, he/she received 3 shuffled cards, each containing an item of the sequence. The cards were given to the child in a pile and their task was to put them on the table in the order of presentation. Irrespective of whether the response was correct, the next trial was indicated with picture of a rocket and applause. To find a good balance between the duration of the experiment (adapting it to the attentional span of young children), and the amount of trials for the calculation of the variables of interest, 12 trials per condition were collected. As such, 72 trials per participant were collected for the evaluation of the OPE.

==== INSERT FIGURE 1 HERE =====

Analyses

The analyses were conducted in two parts. First, we verified whether at the group-level, there is evidence for the presence of an OPE. For this purpose, a repeated measures ANOVA with *Ordinal position* in WM (3 levels: 1 to 3) and *Response location* (2 levels: left, right) was

performed, complemented with the regression linear approach described by Lorch and Myers (ref. 20, see also ref. 21). This method consists of computing the difference in RTs (dRT; RT right side minus RT left side) for each position in WM (from 1 to 3) separately. For each participant, these values were entered in a regression analysis with *Ordinal position* as predictor. With this approach, a negative regression weight reflects faster left-sided responses for early items of the memorized sequence and faster right-sided responses for late items of the sequence (and thus a left-to-right mapping of serial order to space). A t-test was used to evaluate whether the regression weights of the group deviated significantly from zero.

To shed a further light on the prevalence of the OPE at the individual level, a second series of analyses was conducted. To determine whether an individual child showed a reliable OPE, we used the bootstrapping technique that was recently developed by Cipora and colleagues (ref. 19; for the R-script see: <https://osf.io/n7szg/>). The logic behind this technique is the following: 5000 bootstrap samples (randomly sampled with replacement, each containing 12 trials per condition) were generated from the dataset of each individual subject. From each generated sample, the OPE was calculated using the regression approach described above. Based on the 5000 obtained regression weights, the values corresponding with the 10th and 90th percentile were used to determine the confidence interval (i.e. the 80% CI) around the empirically observed OPE. An individual's OPE was considered to be reliable, if this confidence interval did not contain the value 'zero'. In addition, to ensure that the observed reliable OPE's were not due to chance, we also calculated the 'H0 confidence intervals' (see ref. 19 for more details). Here, random sampling with replacement was used for each individual dataset to randomly allocate trials from the different "*Ordinal position*" conditions to two sets: one for 'virtual' right responses and one for 'virtual' left responses (12 trials per condition). Subsequently for each of these samples, the slope of the OPE was calculated. Because of the random allocation of the response location per bootstrap sample, the expected average is around 0. Next confidence intervals were determined around this average, and it was verified whether the empirically observed OPE slope fell within this interval or not. If this value fell outside this interval, it was considered to be reliable. It is known that bootstrapping can be sensitive to outliers in the data²³. To ensure that the observed reliable OPE's are not caused by potential outliers, we (1) applied a conservative data-trimming procedure, and (2) calculated the split-half correlation coefficient (with Spearman-Brown correction) of the group of children who reliably map serial order to space.

RESULTS

Only correctly responded probe trials from blocks where the verification task was correct, were included. The average amount of correctly recalled sequences was 11.64 (SD=.88), and from these sequences, the average proportion of correctly identified probes was 99% (SD=1%).

Reaction times outside the 250-4000ms interval were considered as non-responses (3% (SD=4%) of the data fell outside this interval). Next, only RTs that fell within the range of ± 2 SD from the average RT of the experimental condition to which the trial belonged were considered. Using this procedure, 5% (SD=2%) of the trials were excluded. After these trimming procedures, on average 10.61 trials per condition (SD=1.15) were maintained. The overall RT was 1454ms (SD=298ms) and the overall accuracy 88% (SD=8%). There was no indication of a speed-accuracy trade-off as the correlation between RT and accuracy was $r=-.36$.

The repeated measures ANOVA with *Ordinal position* (3 levels) and *Response location* (2 levels) did not reveal a OPE at the group level, as the interaction between both factors was far from significant [$F(2, 158)=.26, p=.77$]. Also, in the regression analysis there was no significant OPE at the group level. The average slope was 11.52 (SD=174.42) [$t(79)=.59, p=.56$].

The analyses of the individual consistency on the other hand, revealed that in 36% (i.e. 29 out of 80) of the children a reliable OPE was observed, as for them, the bootstrapped confidence intervals around the OPE did not contain 0. Importantly, this result is not caused by (remaining) outliers in the data, as the split-half reliability of the OPE in these 29 children was $r=.75$. Interestingly, when taking a closer look at those showing a reliable OPE, 16 of the 29 children mapped the WM items in a left-to-right fashion, while 13 of the 29 children mapped the items from right-to-left (see Figure 2). The results of the H0-bootstrapping revealed virtually the same pattern of results (97.5% correspondence), indicating that the observed proportions were not due to random fluctuations in the data. With this analysis 34% (27 of the 80) children showed a reliable OPE, 14 of them in a left-to-right orientation and 13 of them a right-to-left orientation.

==== INSERT FIGURE 2 HERE =====

DISCUSSION

Despite an absence of the OPE at the group level, the current study showed that about one out of every three pre-literate children demonstrated a reliable OPE at the individual level. Interestingly, in this subgroup with a reliable OPE, the orientation was more or less equally divided between coding from left-to-right and from right-to-left. Where the overall proportion of participants showing a reliable OPE is comparable to the proportion observed in adults (see ref. 19 for a discussion how consistent group-effects in adults can be driven by a minority), the equal proportion of left-to-right and right-to-left mappers is not found in adults, as the large majority (about 87.5%) of the adults who show a reliable OPE, map order according to the reading habits (Appendix 1). As such, the current study demonstrated for the first time that spontaneous spatial coding of serial order in WM can be observed in young children before they obtained formal literacy education, and that the orientation of this coding is not yet fully aligned to the reading direction of their dominant language in half of the children. This suggests that the reuse of spatial machinery for serial order coding across non-spatially defined items in WM is not uniquely evoked by literacy (but it may be shaped by it; see below). It is important to note that 81 of the tested children spoke a left-to-right language at home and only two a right-to-left language (and one of these two showing a reliable left-to-right OPE; for 1 child the information about the home language was missing). Crucially, the observed individual differences in orientation (left-to-right versus right-to-left) thus rule out a critical role for either pretend reading and writing (stimulated at school or at home), or more rudimentary reading and writing skills that precede literacy²⁴ as a driving factor underlying the OPE because these, too, can be assumed to generally follow the culturally determined reading habits.

One may argue that already before us, a study by Bulf et al.²⁵ presented indications that spatial coding is involved in serial order mechanisms before literacy is acquired. They observed that 7-month-olds infants possess the ability to learn (and transfer) rule-like patterns from visual sequences of geometric shapes when these were presented from left-to-right, but not when presented from right-to-left – suggesting that ordinal information is preferentially coded in a left-to-right fashion already at this young age. We believe, however, that these findings relate to a mechanism that is qualitatively different from serial order WM which we investigated here with the OPE. Our main argument is that the OPE is based on an allocentric coding of space^{6,26,27}, whereas spatial competence in terms of allocentric codes only emerges around 22 months of age²⁸. Moreover, in previous work we have framed the OPE in terms (hippocampal-dependent) item-space binding (i.e., coding serial order across items by binding the items to systematically arranged, spatially defined position markers^{15,16}), a mechanism recycled from

the brain's systems for mnemonic coding of external space. Importantly, it is known that arbitrary bindings that would be required for the OPE, are enabled only from the beginning of the third year of life in these external spatial systems ²⁹. Overall, rather than relating to the flexible use of spontaneous spatial coding for serial order ability in WM, we believe that the above findings with 7-month-old infants are a reflection of left-to-right biases that are evoked by biological factors such as brain lateralization in the control of spatial attention ^{30,31}. Whereas the left-to-right oriented spatial coding of serial order observed in some children of the current study could potentially be explained by the lateralization in the brain's attentional systems, the latter could not explain the right-to-left oriented coding observed here in a different subset of children.

The individual differences in orientation of spatial coding for serial order – and the resulting absence of an OPE at the group level – that we observed in the current study, imply a need for caution in the interpretation of absent OPEs at the group level ³². Future studies will have to explore the prevalence of the OPE at the individual level before interpreting an absent OPE at the group level as support for a lack of systematic spatial coding. Moreover, previous studies that reported on absent OPEs at the group level (e.g. ref. 9,11) need to be interpreted with this limitation in mind as well (see e.g. Appendix 1). Side-stepping from the current paper, a similar argument could be made for only exploring spatial coding along the horizontal axis – since vertical coding may also be a possibility (e.g. ref. ³³).

Finally, we present a tentative working hypothesis to inspire future studies in understanding spatial coding for serial order ability – and its individual differences. We postulate that recycling the evolutionary empowered spatial machinery for serial order has proven to be highly effective (cf. neural recycling; ref. ³⁴). Temporarily binding items to coordinates on an allocentric reference frame allows for flexible coding of serial order ²⁶, and for spatial attention to contribute to critical WM processes such as retrieval, refreshing, and scanning ¹². The availability of this recycled system underlies the OPE that was observed for a subset of preliterate children in the current study. The large individual differences in preferred orientation (e.g., left-to-right versus right-to-left) of coding relates to the flexibility that allocentric coding allows for. As long as conventions from literacy have not yet drilled the system into consistent spatial coding across individuals, within-subject systematicity in orientation will be largely determined by – for example – visuo-motor experiences (e.g., finger counting, hand dominance, counting directions while pointing objects, organizing objects in rows or piles) that come with large individual differences ^{35–37}. Systematicity in orientation may

potentially even lack *within* the preliterate child – in line with the idea that at a young age many intra-individual differences in motor tasks exist ^{38,39}. Interestingly, adults lacking literacy conventions also demonstrate more variable orientations in serial order processes ¹⁸. Once literacy is being acquired, it shapes the orientation of spontaneous spatial coding in WM such that the orientation normalizes across individuals. This reading-based normalization accounts for the robust group-level OPE in adults (see above), and also explains why Guida et al. ¹¹ observed an OPE in adults that was oriented according to reading direction. Interesting to note, when this orientation normalization is “applied” to our dataset (i.e. changing the sign of the OPE of those children with a reliable right-to-left OPE, see 40 for a similar application), a significant effect at the group level emerges.

We acknowledge that our working hypothesis regarding the developmental trajectory of the spatial coding of serial order and its underlying mechanisms remains highly speculative for now. Future studies are required for further testing of its premises. One straightforward prediction is that in a longitudinal design, it will be the right-to-left coding children who will turn into left-to-right coders when becoming skilled in reading/writing. Moreover, our hypothesis is clearly incomplete in terms of understanding the individual differences in the mechanisms underlying serial order WM. For example, whereas about 36% of the preliterate children in the current study demonstrated a reliable OPE, still 64% of the tested children did not show such a systematic OPE and yet were perfectly capable of solving the serial order task. We can think of several reasons for the latter observation. First, spatial coding was involved in their serial order processes but with large within-subject variability (e.g., interchangeably using left-to-right and right-to-left coding across trials). Second, two complementary systems for serial order of non-spatially defined items have been proposed, the phonological loop and an attentional refreshing mechanism ⁴¹. Whereas spatial coding is probably linked to the latter attentional mechanism ^{9,15}, it could be that the majority of preliterate children in the current study solved the task via the phonological loop (thus bypassing spatial coding). Third, besides space, other potential position marker systems for the coding of serial order in WM have been proposed (a.o. time signals ³ and magnitude codes ⁴), although their existence have mainly been proposed on the basis of computational modelling (i.e. without direct empirical evidence), it is possible that those children who do not show a OPE rely upon one of those mechanisms. Finally, the task that we used to test the OPE in children may be lacking in terms of sensitivity to detect systematic spatial coding across all individuals. Further research is needed to validate

these potential explanations and to see whether their explanatory power remain constant over development.

Conclusion

Overall, the current study provides novel insight into the developmental nature of serial order coding in verbal WM: about 40% of the tested preliterate children already systematically associated serial stored items in WM to space, half of them from left-to-right, the other half from right-to-left. These observations clearly show that this spatial coding is not a side-effect of our ability to read and write. As an alternative, we postulate that this spatial coding has its origin in different visuo-motor experiences that are gradually build up and ‘calibrates’ in orientation once literacy is acquired. As a final note, the current study also demonstrates the potential fallacies of determining absence or presence of the OPE at the group level. Even when absent at the group level, reliable effects can be observed at the individual level, which may be informative for further theoretical developments.

AUTHOR CONTRIBUTIONS

W.F. and J.-P.v.D. (in alphabetical order) developed the study concept and contributed to the study design. The experiment was programmed by J.-P.v.D. and data collection was performed under his supervision. J.-P.v.D. conducted the Data analyses. E.A., W.F., and J.-P.v.D. (in alphabetical order) contributed to the theoretical framework. J.-P.v.D. and E.A. drafted the final manuscript, and E.A., W.F., and J.-P.v.D. (in alphabetical order) provided critical revisions and approved it. J.-P.v.D. accepts responsibility for the integrity of the data analyzed.

ACKNOWLEDGEMENTS

We would like to thank Yasmine Saelen, Ada Desyn, Laura De Sloover and Sarah de Wever for their help with the preparation of the study and their devoted contribution to the data-collection.

COMPETING INTERESTS

The authors declare no competing interests.

REFERENCES

1. Marshuetz, C. Order information in working memory: An integrative review of evidence from brain and behavior. *Psychol. Bull.* **131**, 323–339 (2005).
2. Hurlstone, M. J., Hitch, G. J. & Baddeley, A. D. Memory for serial order across domains: An overview of the literature and directions for future research. *Psychol. Bull.* **140**, 339–373 (2014).
3. Brown, G. D. A., Hulme, C. & Preece, T. Oscillator-Based Memory for Serial Order. *Psychol. Rev.* **107**, 127–181 (2000).
4. Botvinick, M. & Watanabe, T. From numerosity to ordinal rank: A gain-field model of serial order representation in cortical working memory. *J. Neurosci.* **27**, 8636–8642 (2007).
5. van Dijck, J. P. & Fias, W. A working memory account for spatial-numerical associations. *Cognition* **119**, 114–119 (2011).
6. Bottini, R., Mattioni, S. & Collignon, O. Early blindness alters the spatial organization of verbal working memory. *Cortex* **83**, 271–279 (2016).
7. De Belder, M., Abrahamse, E., Kerckhof, M., Fias, W. & Van Dijck, J.-P. Serial Position Markers in Space: Visuospatial Priming of Serial Order Working Memory Retrieval. *journals.plos.org* **10**, (2015).
8. Ginsburg, V. & Gevers, W. Spatial coding of ordinal information in short- and long-term memory. *Front. Hum. Neurosci.* **9**, 8 (2015).
9. Ginsburg, V., Archambeau, K., van Dijck, J. P., Chetail, F. & Gevers, W. Coding of serial order in verbal, visual and spatial working memory. *J. Exp. Psychol. Gen.* **146**, 632–650 (2017).
10. Guida, A., Leroux, A., Lavielle-Guida, M. & Noël, Y. A SPoARC in the Dark: Spatialization in Verbal Immediate Memory. *Cogn. Sci.* **40**, 2108–2121 (2016).
11. Guida, A. *et al.* Spatialization in working memory is related to literacy and reading direction: Culture “literarily” directs our thoughts. *Cognition* **175**, 96–100 (2018).

12. van Dijck, J. P., Abrahamse, E. L., Majerus, S. & Fias, W. Spatial Attention Interacts With Serial-Order Retrieval From Verbal Working Memory. *Psychol. Sci.* **24**, 1854–1859 (2013).
13. van Dijck, J.-P., Abrahamse, E. L., Acar, F., Ketels, B. & Fias, W. A Working Memory Account of the Interaction between Numbers and Spatial Attention. *Q. J. Exp. Psychol.* **67**, 1500–1513 (2014).
14. Rinaldi, L., Brugger, P., Bockisch, C. J., Bertolini, G. & Girelli, L. Keeping an eye on serial order: Ocular movements bind space and time. *Cognition* **142**, 291–298 (2015).
15. Abrahamse, E. L., van Dijck, J. P. & Fias, W. Grounding Verbal Working Memory: The Case of Serial Order. *Curr. Dir. Psychol. Sci.* **26**, 429–433 (2017).
16. Abrahamse, E., van Dijck, J. P., Majerus, S. & Fias, W. Finding the answer in space: The mental whiteboard hypothesis on serial order in working memory. *Front. Hum. Neurosci.* **8**, 932 (2014).
17. Lewkowicz, D. J. Perception of serial order in infants. *Dev. Sci.* **7**, 175–184 (2004).
18. Cooperrider, K., Marghetis, T. & Núñez, R. Where Does the Ordered Line Come From? Evidence From a Culture of Papua New Guinea. *Psychol. Sci.* **28**, 599–608 (2017).
19. Cipora, K. *et al.* A Minority pulls the sample mean: on the individual prevalence of robust group-level cognitive phenomena – the instance of the SNARC effect. *PsyArXiv* (2019) doi:10.31234/OSF.IO/BWYR3.
20. Ghyselinck, M., Custers, R. & Brysbaert, M. Age_of_Acquisition ratings for 2332 Dutch words from 49 different semantic categories. *Psychol. Belg.* **43**, 181–214 (2003).
21. Lorch, R. F. & Myers, J. L. Regression Analyses of Repeated Measures Data in Cognitive Research. *J. Exp. Psychol. Learn. Mem. Cogn.* **16**, 149–157 (1990).
22. Fias, W., Brysbaert, M., Geypens, F. & D’Ydewalle, G. The importance of magnitude information in number processing: Evidence from the SNARC effect. *Math. Cogn.* **2**, 95–110 (1996).

23. Rousselet, G., Cyril, P. & Wilcox, R. R. A practical introduction to the bootstrap: a versatile method to make inferences by using data-driven simulations. *PsyArXiv* (2019) doi:10.17605/OSF.IO/8B4T5.
24. Patro, K., Nuerk, H. C. & Cress, U. Mental Number Line in the Preliterate Brain: The Role of Early Directional Experiences. *Child Dev. Perspect.* **10**, 172–177 (2016).
25. Bulf, H., De Hevia, M. D., Gariboldi, V. & Cassia, V. M. Infants learn better from left to right: A directional bias in infants' sequence learning. *Sci. Rep.* **7**, 1–6 (2017).
26. Guida, A., Abrahamse, E. L. & Van Dijck, J.-P. About the interplay between internal and external spatial codes in the mind : implications for serial order. *Ann. N. Y. Acad. Sci.* doi:10.1111/nyas.14341.
27. Wang, Q., Nie, T., Zhang, W. & Shi, W. The Mechanism of the Ordinal Position Effect: Stability Across Sense Modalities and the Hands Crossed Context. *Iperception.* **10**, 204166951984107 (2019).
28. Newcombe, N., Huttenlocher, J., Drummey, A. B. & Wiley, J. G. The development of spatial location coding: Place learning and dead reckoning in the second and third years. *Cogn. Dev.* **13**, 185–200 (1998).
29. Ghetti, S. Development of item–space and item–time binding. *Curr. Opin. Behav. Sci.* **17**, 211–216 (2017).
30. Patro, K. & Nuerk, H.-C. Limitations of Trans-Species Inferences: The Case of Spatial-Numerical Associations in Chicks and Humans. *Cogn. Sci.* **41**, 2267–2274 (2017).
31. Felisatti, A., Laubrock, J., Shaki, S. & Fischer, M. H. Commentary: A mental number line in human newborns. *Front. Hum. Neurosci.* **14**, 99 (2020).
32. Abrahamse, E. & Guida, A. Commentary: Coding of serial order in verbal, visual and spatial working memory. *Front. Psychol.* **9**, 2330 (2018).
33. Zhou, D. *et al.* The metaphoric nature of the ordinal position effect. *Q. J. Exp. Psychol. (Hove).* **72**, 2121–2129 (2019).

34. Anderson, M. L. Neural reuse: A fundamental organizational principle of the brain. *Behavioral and Brain Sciences* vol. 33 245–266 (2010).
35. Morrissey, K., Hallett, D., Wynes, R., Kang, J. & Han, M. Finger-counting habits, not finger movements, predict simple arithmetic problem solving. *Psychol. Res.* **84**, 140–151 (2020).
36. Previtali, P., Rinaldi, L. & Girelli, L. Nature or Nurture in Finger Counting: A Review on the Determinants of the Direction of Number?Finger Mapping. *Front. Psychol.* **2**, 363 (2011).
37. Knudsen, B., Fischer, M. H. & Aschersleben, G. Development of spatial preferences for counting and picture naming. *Psychol. Res.* **79**, 939–949 (2014).
38. Adolph, K. E., Cole, W. G. & Vereijken, B. Intraindividual variability in the development of motor skills in childhood. in *Handbook of Intraindividual Variability Across the Life Span* (eds. Diehl, M., Hooker, K. & Silwinski, M.) 59–83 (Routledge/Taylor & Francis Group, 2014). doi:10.4324/9780203113066.
39. Rivera, D. *et al.* Intraindividual Variability Measurement of Fine Manual Motor Skills in Children Using an Electronic Pegboard: Cohort Study. *JMIR mHealth uHealth* **7**, e12434 (2019).
40. Kramer, P., Bressan, P. & Grassi, M. The SNARC effect is associated with worse mathematical intelligence and poorer time estimation. *R. Soc. Open Sci.* **5**, (2018).
41. Camos, V. Storing Verbal Information in Working Memory. *Curr. Dir. Psychol. Sci.* **24**, 440–445 (2015).
42. Demoulin, C. & Kolinsky, R. Does learning to read shape verbal working memory? *Psychonomic Bulletin and Review* vol. 23 703–722 (2016).

FIGURE CAPTIONS

Figure 1: The three phases of each experimental block. The experiment consists of 12 such blocks.

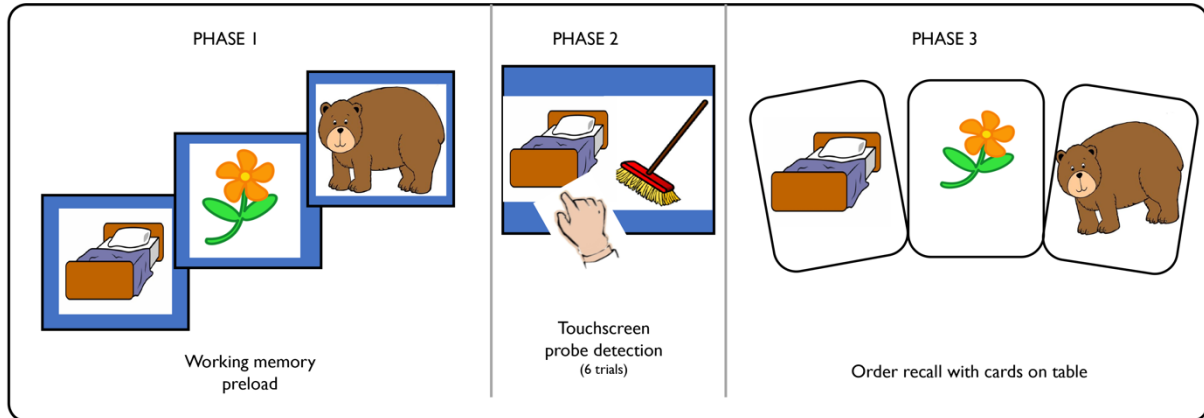
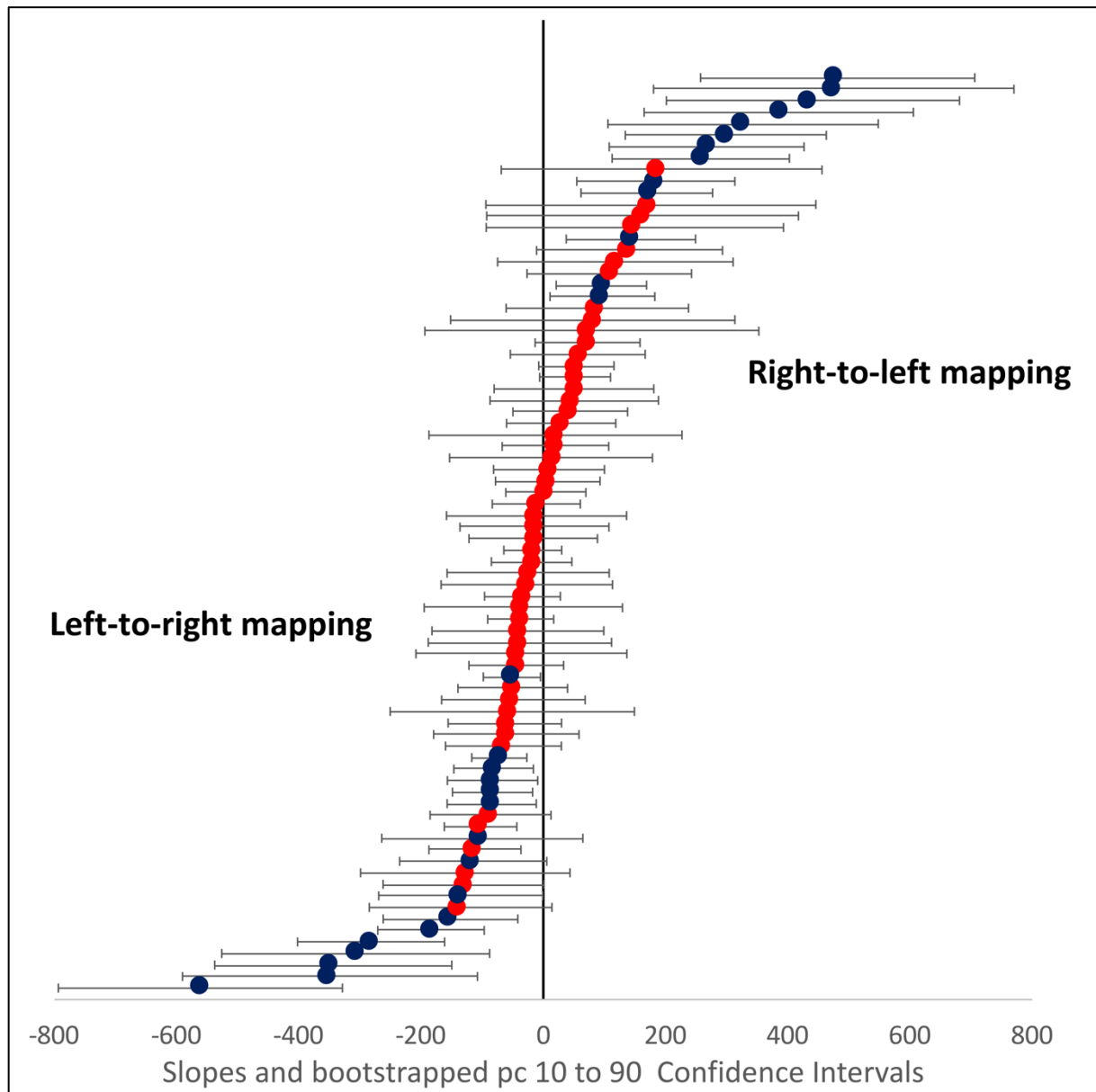


Figure 2: The results of the individual consistency analyses.

The dots represent the empirically observed regression slopes of each individual and reflect the OPE. The lines around the dots are the 10 to 90 percentile confidence intervals. The confidence intervals are obtained from the 5000 bootstrap samples. The dots in blue indicate the presence of a reliable OPE. When those dots are below zero, the child shows a reliable left-to-right mapping of serial order in WM. When the dots have a positive value, a reliable right-to-left mapping is observed.



Appendix 1

Proportion of reliable individual OPEs in adults with different reading directions

Data set	Task	Country	Reading direction	N° of participants	Trials per condition	Left-right coders (%/number)	Inconsistent coders (%/number)	Right-left coders (%/number)	Total spatial coding (%/number)
Guida et al. (2018; ref. 11)	Probe-detection	Egypt	right-to-left	19	32	16% / 3	52% / 10	32% / 6	47% / 9
	Probe-detection	Egypt	illiterate	20	32	10% / 2	75% / 15	15% / 3	25% / 5
	Probe-detection	Belgium & France	left-to-right	20	32	30% / 6	65% / 13	5% / 1	35% / 7
Razoulzadeh et al. (in prep.)	Fruit/vegetable categorization	Iran	right-to-left	23	21	4% / 1	61% / 14	35% / 8	39% / 9
van Dijck et al. (2013; ref. 12)	Dot-detection	Belgium	left-to-right	19	50	47% / 9	47% / 9	5% / 1	53% / 10
van Dijck & Fias (2011; ref. 13)	Fruit/vegetable categorization	Belgium	left-to-right	36	20	56% / 20	42% / 15	3% / 1	58% / 21
Summary						Spatial coding according to reading habits: 87.5% / 49			Average proportion spatial coding: 45% / 61